BOOK MARKET ANALYTICS

Books to Scrape catalogue: pricing, ratings and genre trends

Assignment	Data Science 500
Prepared	11 October 2026
Student name	[Enter your full name]
Student ITS number	[Enter your ITS number]
Qualification / year / semester	[Complete before submission]

End-to-end web scraping, exploratory analysis and predictive modelling

Declaration of Originality

I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

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1. Executive summary

This study scraped 1,000 unique product listings across 50 genres from Books to Scrape. The average listed price was £35.07 (median £35.98; range £10.00–£59.99). A linear regression model using genre, star rating, availability, stock quantity and title length achieved test MAE £12.91, RMSE £15.00 and R^2 -0.071. It performed slightly worse than a simple mean-price baseline, so the selected features do not provide useful predictive improvement in this split. The site is a scraping sandbox; these findings are illustrative, not representative of the commercial book market.

2. Introduction

Retailers and publishers need evidence about how catalogue attributes relate to list prices when planning assortment and merchandising. This project investigates price patterns across book genres and star ratings and evaluates whether available listing features can estimate price. The objective is to demonstrate a reproducible data-science workflow, not to produce a deployable pricing system.

3. Data acquisition and methodology

The data were collected on 11 October 2026 from the Books to Scrape demonstration site (Books to Scrape, n.d.). Python requests retrieved catalogue and genre listing pages; BeautifulSoup parsed product cards using CSS selectors (Richardson, n.d.). The scraper followed pagination across the catalogue and genre pages, mapped product URLs to genres, checked HTTP responses, and paused briefly between requests. For each book it captured title, GBP price, rating word, raw stock availability, normalized stock state, quantity when displayed, genre and product URL.

4. Data cleaning and preparation

Duplicate product URLs were removed. Prices were converted to numeric GBP and rating words One–Five mapped to integers 1–5. Rows with missing or invalid price/rating were excluded from the modelling data; blank genres were retained as Unknown. Stock quantity was parsed where present, missing quantities were replaced by zero, and a separate missingness indicator was retained. Derived predictors included an in-stock indicator and title character length. The resulting 1,000-row dataset is included as CSV.

5. Exploratory analysis and findings

The observed price range was £10.00–£59.99; mean £35.07, median £35.98, and standard deviation £14.45. The mean star rating was 2.92/5. All 100% of listings were marked in stock at collection, so the binary stock feature has no variation and cannot distinguish prices in this snapshot.

Among genres with at least 10 books, the highest mean prices were: Travel: £39.79 (n=11); Fantasy: £39.59 (n=48); History: £37.29 (n=18); Womens Fiction: £36.79 (n=17); Classics: £36.55 (n=19).

Among genres with at least 10 books, the lowest mean prices were: Food and Drink: £31.41 (n=30); Thriller: £31.43 (n=11); Mystery: £31.72 (n=32); Business: £32.46 (n=12); Childrens: £32.64 (n=29).

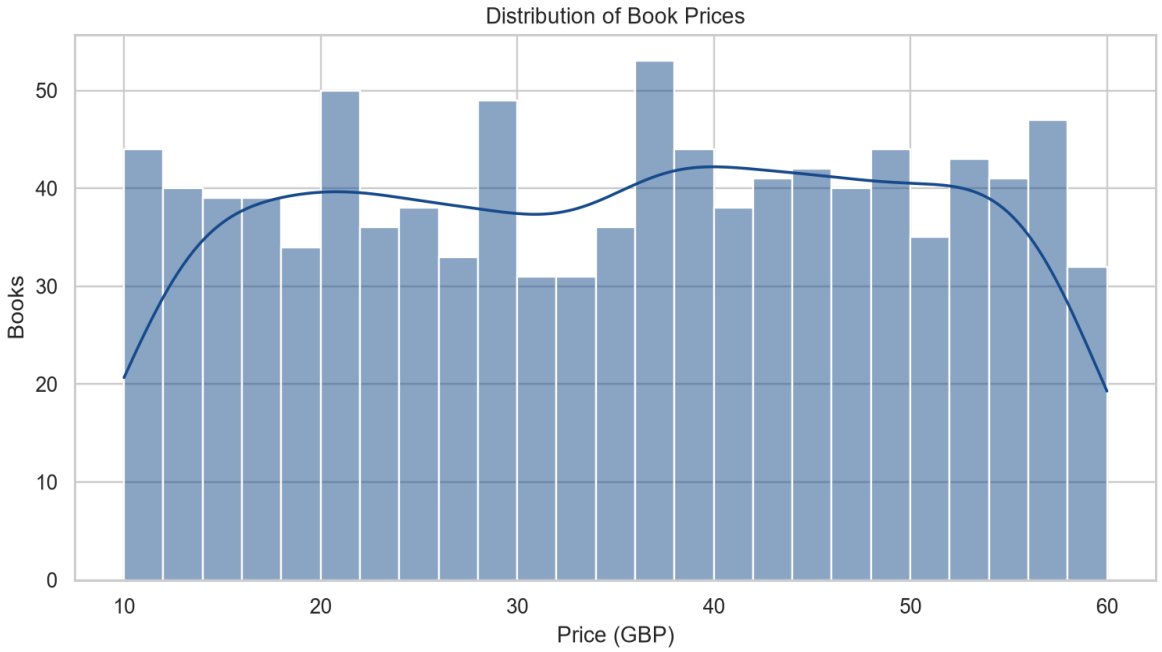


Figure 1. Distribution of listed prices in GBP.

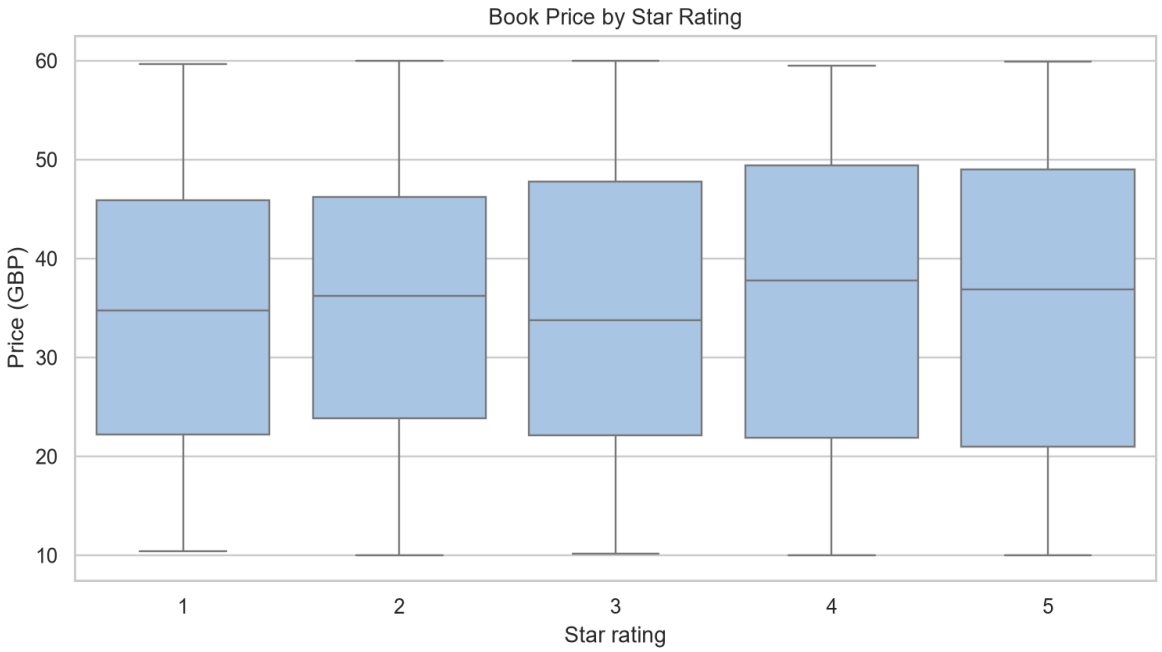


Figure 2. Price by star rating; distributions overlap substantially.

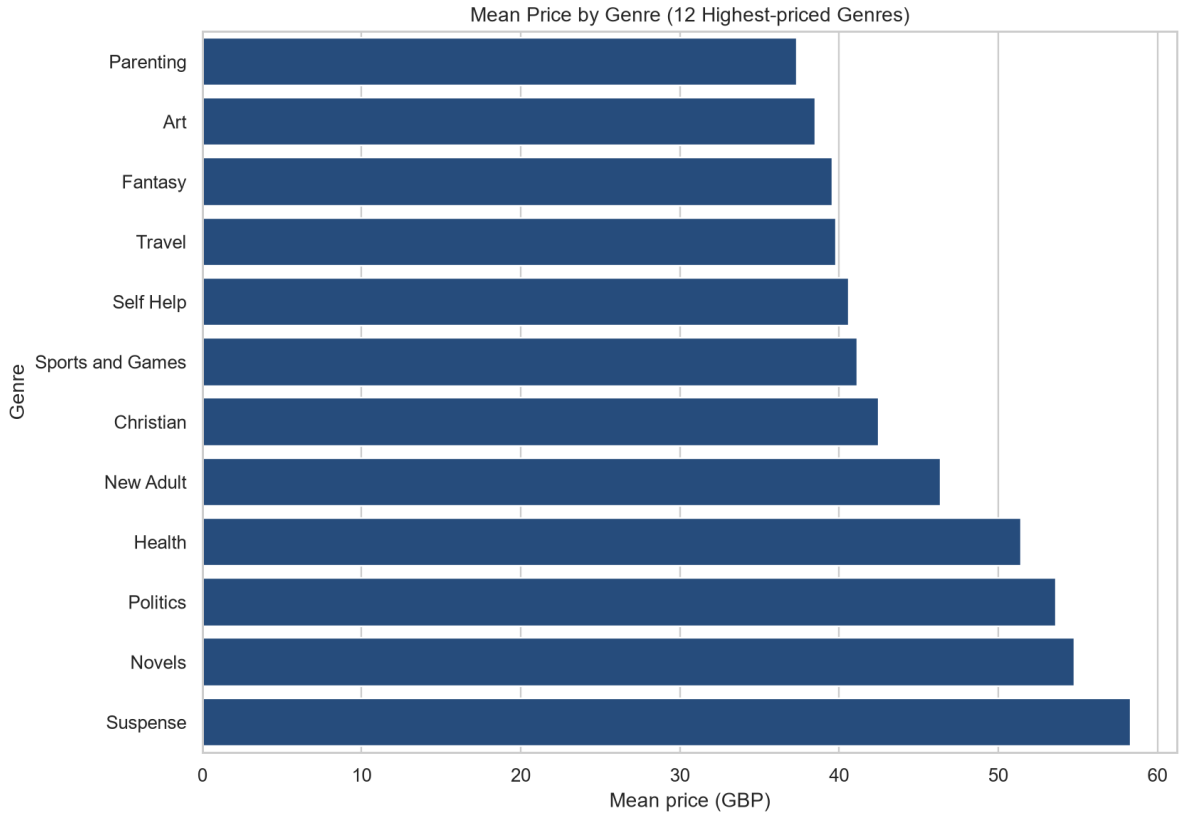


Figure 3. Mean price in the 12 highest-priced genres in the sample.

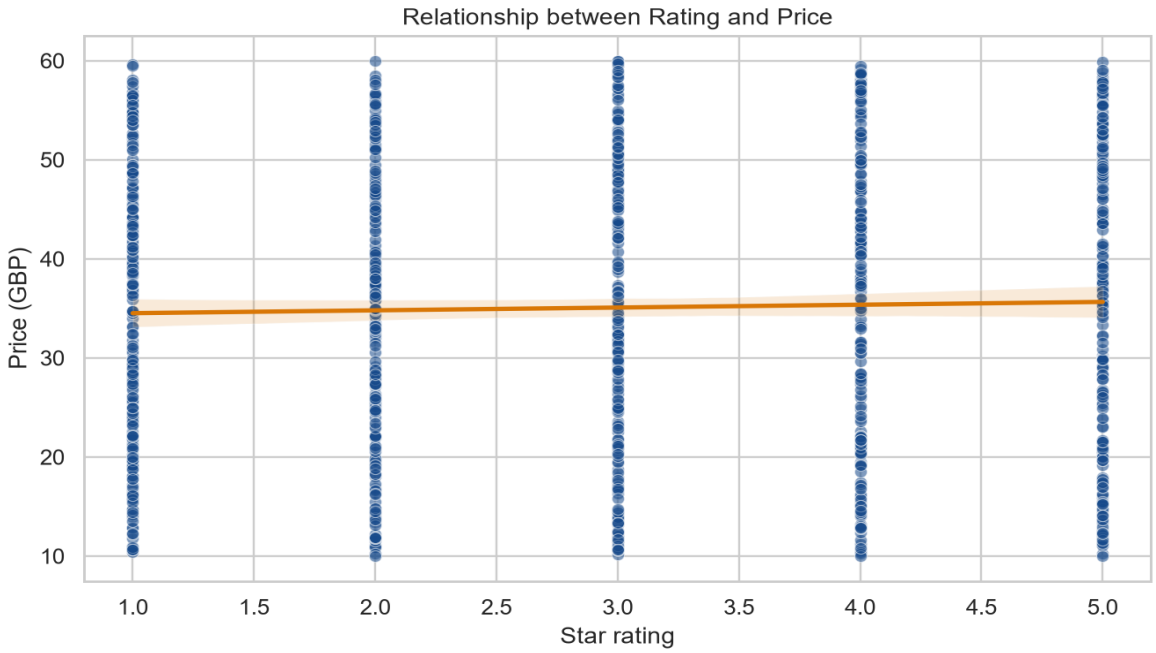


Figure 4. Star rating and price; the association is not causal.

6. Predictive modelling

The target was price_gbp. Predictors were numeric rating, in-stock state, stock quantity, genre and title length. Numeric features were standardized and genre one-hot encoded in a scikit-learn preprocessing pipeline. An 80/20 train/test split used random_state=42; preprocessing was fitted

within the training pipeline to avoid leaking test information. LinearRegression was compared with a mean-price DummyRegressor (scikit-learn developers, n.d.).

Metric	Linear regression	Mean baseline
MAE (£)	12.911	12.662
RMSE (£)	15.002	14.530
R-squared	-0.071	-0.005

MAE is mean absolute error, RMSE penalizes larger errors, and R^2 measures test-set variance explained relative to a mean predictor. The model's negative R^2 and higher MAE/RMSE show that it did not outperform the baseline on the held-out sample. This is an important negative result: the chosen listing features provide little reliable price signal in this dataset.

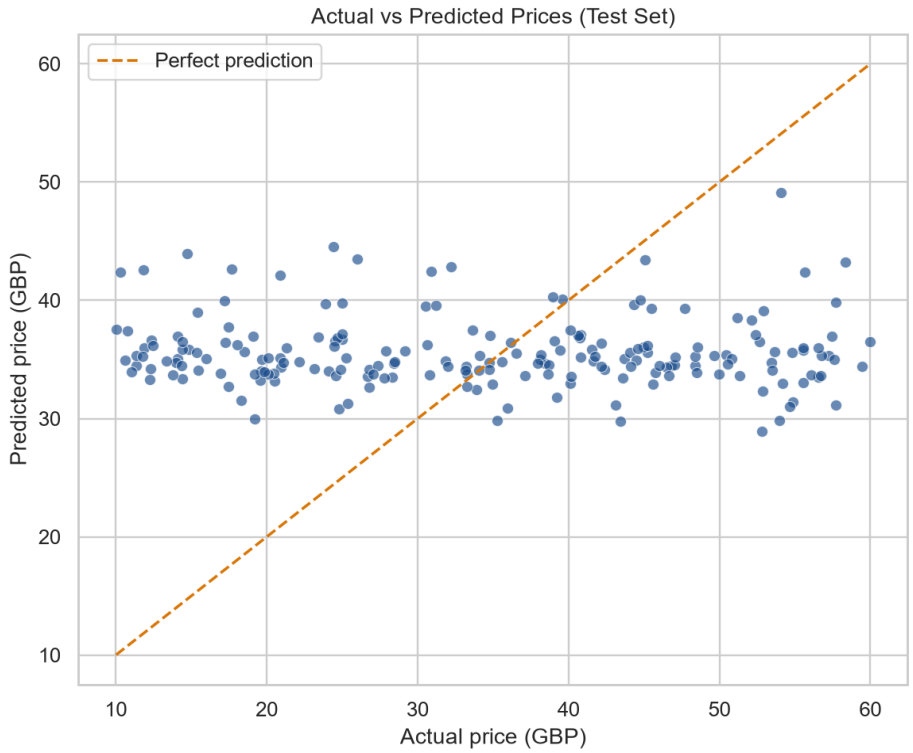


Figure 5. Held-out actual versus predicted prices.

7. Discussion and recommendations

Several limitations constrain interpretation. Books to Scrape is an intentionally scrapeable demonstration catalogue, not a random sample of commercial listings. Every listing was in stock at the time of collection, leaving no stock-state variation. Some genres have small counts; genre averages should be compared only with sample size in view. The model is linear and excludes potentially important factors such as edition, format, publisher, page count, publication date, promotions and demand. A star rating is not the same as popularity or sales, and this cross-sectional analysis cannot support causal claims.

Recommendations. Retailers should treat genre summaries as exploratory and validate patterns against their own sales, cost and inventory data. Publishers should collect richer edition and format attributes before evaluating price differences. Book buyers should consider price and star rating as separate signals; the scrape has no review counts or sales data. Future work should use repeated snapshots and real retailer data, add richer features, apply cross-validation, compare regularized and nonlinear models, and report errors by genre.

8. Conclusion

The assignment produced a reproducible scrape, a cleaned dataset of 1,000 books across 50 genres, visual exploration and a held-out linear-regression evaluation. Average listed price was £35.07. The model did not beat the mean baseline, so the tested features are insufficient for useful price prediction. This is a useful result for guiding feature collection, but not a commercial pricing recommendation.

References

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